TEAM #16
Smarter Freight, Smarter Routes
Challenge Sponsor: Trans Ontario
Every day, freight carriers make hundreds of decisions about which shipments go on which trucks, how those trucks should be loaded, and the most efficient routes to get everything delivered on time.

But what happens when a shipment doesn't fit, a driver calls in sick, or a delivery needs to be rescheduled?
CHALLENGE STATEMENT
“How can AI and technology help freight carriers plan deliveries more efficiently, optimize truck capacity, and adapt to unexpected disruptions?”
01
The Challenge
Every day, freight carriers make hundreds of decisions about which shipments go on which trucks, how those trucks should be loaded, and the most efficient routes to get everything delivered on time.
But what happens when a shipment doesn't fit, a driver calls in sick, or a delivery needs to be rescheduled?
Your challenge is to explore how technology can make freight planning smarter, faster, and more adaptable.
02
Background & Context
Trans Ontario is a regional freight carrier operating across Ontario and Quebec, with additional shipping services extending across Canada through partner carriers.
The company manages approximately 300–500 shipments per operating day across a fleet of more than 50 vehicles.
Currently, much of the planning process is manual. Dispatchers organize shipping documents, assign freight to vehicles, determine delivery routes, and coordinate changes as unexpected situations arise.
While an existing transportation management system (TMS) records these decisions and communicates routes to drivers, it does not automatically determine the best plan.
This creates an opportunity to explore how AI, automation, optimization algorithms, and digital tools could improve the way freight is planned and delivered.
03
The Problem
Planning a delivery isn't as simple as finding the shortest route.
Dispatchers must consider several interconnected factors:
Vehicle Capacity: Each truck has different weight, space, and equipment limitations.
Loading Sequence: Freight must be loaded in an order that allows deliveries to be unloaded efficiently.
Delivery Requirements: Customers have different delivery windows, unloading needs, and location restrictions.

Driver Availability: Drivers must have the appropriate licences and remain within legal working-hour limits.
Carrier Partnerships: Some shipments may need to be transferred to partner carriers rather than delivered directly.
Unexpected Disruptions: Breakdowns, cancellations, delays, and last-minute shipments can require an entire plan to change.
These decisions are connected. Changing one delivery route could affect how a truck is loaded, which shipments it can carry, and whether other deliveries arrive on time.
The challenge is to find ways to make these complex decisions more efficient while maintaining safety, reliability, and operational feasibility.

04

Your Mission

Design and prototype an intelligent freight-planning solution that helps dispatchers make better decisions.
Your solution should demonstrate how technology can support three key activities:

1. Plan the Day

Explore how shipments could be assigned to appropriate vehicles, organized into feasible loads, and routed efficiently while considering delivery commitments and operational constraints.

2. Identify Potential Problems

Help dispatchers recognize conflicts before they become costly mistakes, such as overloaded trucks, incompatible equipment, missing shipment information, or delivery windows that cannot be met.

3. Adapt When Things Change

Demonstrate how your solution would respond to an unexpected disruption, such as a truck breakdown, driver absence, late shipment, or failed delivery.
You have the freedom to explore different technical approaches, including AI, optimization algorithms, automation, predictive analytics, and interactive planning tools.

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Requirements & Considerations

Your solution should account for the following:
Safety & Compliance: Vehicle capacity, legal weight limits, driver qualifications, working hours, cargo securement, and applicable transportation regulations must be respected.
Physical Loading: Shipment dimensions, stacking restrictions, equipment compatibility, and unloading order affect whether a proposed route is feasible.
Delivery Commitments: Delivery windows, appointments, and service promises must be considered.
Connected Decisions: Vehicle assignment, load sequence, routing, and partner-carrier decisions must work together rather than being treated as unrelated problems.
Changing Conditions: Your solution must demonstrate how a plan can be adjusted when an unexpected event occurs.
Data Transparency: Missing or estimated information should be clearly identified. Your solution should explain important recommendations rather than making hidden assumptions.
You may simplify certain aspects of the problem for your prototype, provided those assumptions are clearly documented and the core operational requirements are respected.

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What to Build

Your team should develop a functional prototype or proof of concept that demonstrates your approach to the challenge.

Possible solutions include:

An AI-assisted freight planning and optimization tool

An intelligent dispatch dashboard

A truck-loading and route visualization platform

An automated shipment assignment and conflict detection system

A real-time replanning tool that recommends alternatives when disruptions occur

These are examples, not prescribed solutions. You're encouraged to explore your own approach.

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What Success Looks Like

A strong solution should demonstrate how it could help Trans Ontario:

Improve Delivery Reliability: Meet more delivery commitments and identify potential failures earlier.

Reduce Operational Costs: Minimize unnecessary travel, driver hours, and third-party carrier costs.

Use Fleet Capacity Efficiently: Make better use of available vehicles without compromising safety or delivery commitments.

Respond Faster to Disruptions: Help dispatchers evaluate alternatives and make informed decisions.

Build Trust in Recommendations: Clearly communicate why decisions were made and identify assumptions or missing information.

Solutions will be evaluated against the challenge's operational goals, including delivery performance, cost, vehicle utilization, and the feasibility of proposed plans.

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Supporting Resources

TransOntario Brand Assets

ZIP · 78 KB

Sample Routing Report

XLSX · 3.6 MB

Sample Driver Day

XLSX · 448 KB